Airfoil Flow Prediction: *Where* and *Why* Mamba Blocks Enhance U-Net Architectures

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***Abstract*—A convolutional U-Net (DfpNet, Thuerey et al. [1]) approximates the solution of the Reynolds-averaged Navier–Stokes (RANS) equations around airfoils in a single forward pass, far more efficiently than a numerical solver. Since convolutions are local, whereas the pressure obeys a Poisson equation whose elliptic operator couples the whole domain, we investigate whether a Mamba block, a selective state-space model with cost linear in the number of tokens, improves this network, where it should be placed, and how it compares with self-attention. At the deepest levels it adds nothing, although stacked Mamba blocks can replace the deep convolutional path without a measurable loss. At the highest resolution (64×64) it reduces the relative error on v_y by 18% on unseen airfoils in a pre-registered test; self-attention gives a comparable gain at about 17 times the training cost. Ablations show that most of the gain persists without the SSM, the core of Mamba (a 13% reduction, also confirmed on the test set), in line with MambaOut [2]. With 8 times more training data, the gain halves but persists near the airfoil.**

***Index Terms*—RANS, U-Net, Mamba, SSMs, ablation.**

I. Introduction

Computational fluid dynamics (CFD) predicts fluid flows by solving their governing equations numerically: for a Newtonian fluid, the Navier–Stokes (NS) equations are used. However, at high Reynolds numbers $\mathrm{Re} = Uc/\nu$ (flow speed U, chord c, kinematic viscosity ν), direct numerical simulation of turbulent flows is prohibitive, and the Reynolds-averaged Navier–Stokes (RANS) equations are solved instead to model the mean flow affordably.

A neural network trained on RANS solutions can replace the numerical solver when a design loop requires thousands of solutions, mapping geometry and flow conditions to the solution in a single forward pass. Thuerey et al. [1] showed that a convolutional U-Net, DfpNet, predicts pressure and velocity around airfoils with a combined relative error (Sec. III-B) below 3%. While convolutions operate locally, requiring a U-Net to repeatedly reduce resolution to span the whole domain, pressure is governed by an elliptic Poisson equation, making its local values dependent on the entire flow field. Global mixing mechanisms can overcome this mismatch: self-attention [3] relates all pairs of tokens at quadratic cost, while Mamba [4], a selective state-space model (SSM), propagates information at linear cost, and VMamba [5] extends it to images.

We ask whether a Mamba block improves DfpNet, where in the network it helps, and which of its components is responsible: we insert one residual block at one resolution at a time, then remove parts of it, and confirm the main findings with pre-registered tests on unseen airfoils.

II. Background

A. Reynolds-averaged Navier–Stokes equations

For turbulent flows at high Reynolds number ($\mathrm{Re} \in [0.5, 5] \times 10^6$ in our data), the RANS approach decomposes velocity and pressure into a mean and a fluctuating part, $u_i = \bar{u}_i + u_i'$ and $p = \bar{p} + p'$, with $\overline{u_i'} = 0$; the smoother mean fields can thus be computed on much coarser grids. For a steady, incompressible flow, averaging the continuity and momentum equations yields equations for $\bar{u}_i$ and $\bar{p}$ in which the nonlinear convective term produces the Reynolds stresses $\overline{u_i' u_j'}$, new unknowns that leave the system unclosed. In the training data [1], the system is completed with the Boussinesq hypothesis, which expresses the stresses through a turbulent viscosity, obtained from the one-equation Spalart–Allmaras model [6]. Taking the divergence of the averaged momentum equation and using continuity ($\partial_i \bar{u}_i = 0$), the mean pressure satisfies a Poisson equation (ρ: density; repeated indices are summed)

$$\nabla^2 \bar{p} = -\rho\, \partial_i \partial_j \left(\bar{u}_i \bar{u}_j + \overline{u_i' u_j'} \right), \tag{1}$$

which is elliptic, meaning the pressure at any point depends on the source over the whole domain. The mean fields $\bar{p}$, $\bar{u}_x$, $\bar{u}_y$ are the targets of the network, denoted by p, v_x, v_y below.

B. U-Net and receptive field

A U-Net [7] uses skip connections to concatenate feature maps from the encoder to the decoder. DfpNet [1] (Fig. 1, top) is symmetric, employs seven stride-2 convolutions (128×128 to 1×1, 32 to 256 channels, 2.33 M parameters), and is trained with an L1 loss. Its first layer is a single convolution, so the 64×64 skip features undergo minimal processing; the receptive field grows with depth, from 4 px at 64×64 to 46 px at 8×8, while the deeper stages (79% of the weights) extend it to the entire domain. Thus, the U-Net already has global coverage, but lacks adequate representation capacity at high spatial resolution.

C. Global mixing: attention and selective state-space models

For global mixing, a feature map at spatial resolution $H \times W$ is reshaped into a sequence of $L = HW$ tokens $X \in \mathbb{R}^{L \times d}$. Self-attention [3] evaluates all pairwise relationships via

$$\mathrm{Attn}(X) = \mathrm{softmax}\left(QK^\top / \sqrt{d} \right) V, \tag{2}$$

where Q, K, V are linear projections of X. All locations interact directly, at cost $\mathcal{O}(L^2)$ (1.7×10^7 pairs at 64×64).

Conversely, a state-space model (SSM) maps an input sequence $x(t)$ to an output $y(t)$ through a hidden state $h(t)$ via $h'(t) = Ah(t)+Bx(t)$, $y(t) = Ch(t)+Dx(t)$. Discretized with a step size Δ using a Zero-Order Hold assumption ($\bar{A} = \exp(\Delta A)$, $\bar{B} \approx \Delta B$), the relation reduces to the linear recurrence [8]

$$\mathbf{h}_t = \bar{A}\mathbf{h}_{t-1} + \bar{B}\mathbf{x}_t, \qquad \mathbf{y}_t = C\mathbf{h}_t + D\mathbf{x}_t, \tag{3}$$

where D is a per-channel skip term. Through this formulation, each token influences all later ones at a linear cost $\mathcal{O}(L)$.

Mamba [4] makes the SSM *selective* by conditioning Δ, B and C on the input $\mathbf{x}_t$, so each token controls how much of the state is kept. Moreover, since the recursion can be written using an associative operator, it is computed by a hardware-aware parallel scan rather than token by token.

VMamba [5] brings this to images with the SS2D block (Fig. 1, bottom). An input projection expands the n_c channels into two branches of width E; one passes through a 3×3 depthwise convolution and SiLU, is flattened in four orders (rows and columns, forward and backward) and scanned by four SSMs with separate parameters; their outputs are then mapped back to 2D, summed, normalised by a LayerNorm and gated by the SiLU of the other branch, before an output projection back to n_c channels. MambaOut [2] found that the same block without the SSM (a Gated CNN) suffices for ImageNet classification, but falls short of visual Mamba models on detection and segmentation.

D. Related work

Attention [9], [10] and Mamba blocks [11], [12] have been applied to flow prediction, and to U-Nets [13]; these works report the gain of the complete model, while we ask where the block helps and which part is responsible.

III. Method

A. Data, splits and pipeline

We adopt the dataset from [1], comprising UIUC airfoils simulated via OpenFOAM at angles of attack up to $\approx 22.5°$ on a 128×128 grid. Network inputs include the two freestream velocity components and the binary airfoil mask, while targets (p, v_x, v_y) are normalized using $|\mathbf{v}_\infty|^2$ or $|\mathbf{v}_\infty|$ alongside fixed maximum values. We train on the reduced set (6,400 samples across 1,456 airfoils) and evaluate on the official test set (90 samples, 30 unseen airfoils).

In the repository's default validation split, 390 of the 400 files share airfoils with the training set, thus measuring unseen flow conditions on known geometries. To validate on unseen shapes, we re-split strictly by airfoil (5,998 training / 402 validation files across 1,371 / 85 airfoils), causing the validation $\mathrm{rel}_{\mathrm{all}}$ of baseline B to increase by $1.31\times$ and get closer to the test level (test/validation ratio 1.14, vs. 1.50 with the random split). For DR-M and TR († in Table I), 400 more files, one per training airfoil, were held out as a seen-shape validation: they train on 5,598 files and are compared with B, retrained on the same split.

B. Metrics and composition of the error

As in the official repository, the relative error of sample s and channel $c \in \{p, v_x, v_y\}$ is $\mathrm{rel}_c(s) = \sum_i |o_{c,i} - t_{c,i}| / \sum_i |t_{c,i}|$, where o and t are prediction and target, and i runs over the 128×128 cells. The aggregate metric $\mathrm{rel}_{\mathrm{all}}$ pools all channels via a weighted mean:

$$\mathrm{rel}_{\mathrm{all}}(s) = \sum_c w_c(s)\,\mathrm{rel}_c(s), \quad w_c(s) = \frac{\sum_i |t_{c,i}|}{\sum_{c'}\sum_i |t_{c',i}|}. \tag{4}$$

Validation weights average 0.046 for p, 0.825 for v_x, and 0.129 for v_y: v_x is mostly freestream, so $\mathrm{rel}_{\mathrm{all}} \approx \mathrm{rel}_{v_x}$, and one point gained on v_y moves $\mathrm{rel}_{\mathrm{all}}$ by only 0.13. Since a mean of ratios is dominated by samples with a small target signal (e.g. v_y and p at low angles), we also report the aggregate ratio over all samples and the median sample, plus the v_y error in a 4-pixel wall band.

C. Configurations

Each configuration adds one residual block, $\mathbf{y} = \mathbf{x} + \mathrm{Mixer}(\mathrm{LN}(\mathbf{x}))$, after an encoder layer, feeding both the subsequent layer and the skip connection (Fig. 1); the mixer is SS2D (following the official VMamba implementation, with $E = 2n_c$ and state dimension $N = 16$), self-attention, or scan-free SS2D, depending on the configuration. In addition, zero-initializing the output projection ensures exact identity with baseline B at the start (for matched random seeds).

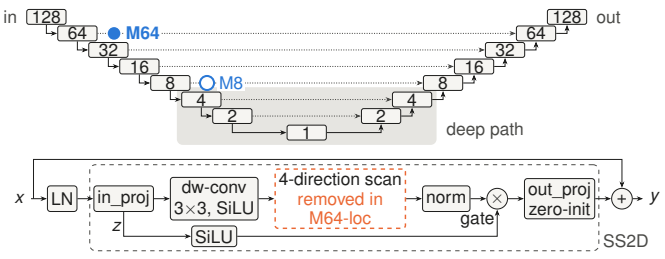


Fig. 1. **Top:** DfpNet with level resolutions and insertion points (M_{64}, M_8; $\mathrm{M}_{64}^{\mathrm{dec}}$ sits in the decoder); the shaded deep path is pruned in TR and replaced by 11 SS2D blocks in DR-M. **Bottom:** residual block with the SS2D mixer; in $\mathrm{M}_{64}^{\mathrm{loc}}$ (M64-loc) the scan is replaced by $D\mathbf{x}$.

Configurations M_l and $\mathrm{M}_{64}^{\mathrm{dec}}$ insert the block at resolution $l \times l$ in the encoder and decoder, respectively; $\mathrm{M}_8^{\mathrm{sh}}$ is an earlier simplification, where the four scans share one complete 1D Mamba layer (projections, gate and its own short convolution, no output norm) and their outputs are summed; SA_{64} uses 4-head self-attention. For deep-path analysis, TR truncates DfpNet below 8×8, while DR-M replaces the pruned layers with 11 SS2D blocks, parameter-matched to B.

To isolate non-sequential effects, $\mathrm{M}_{64}^{\mathrm{loc}}$ replaces each scan path with its skip term $D\mathbf{x}$ (3). This block is close to the local Gated CNN of MambaOut [2], with no spatial propagation beyond 3×3. Five ablations remove one component at a time (gating, output/input norm, expansion, or a dense convolution). As controls, Conv_{64} uses two 3×3 convolutions (similar parameter count, no norm or gating), and B-large widens B to 33 base channels, adding approximately as many parameters

TABLE I

CONFIGURATIONS AND RESULTS. Δ: MEAN OVER SAMPLES OF THE SEED-AVERAGED DIFFERENCE TO B (TEST $\mathrm{rel}_{\mathrm{all}}$ 2.78%, TEST rel_{v_y} 6.64%, VAL. rel_{v_y} 6.07%), IN POINTS, WITH 95% TWO-LEVEL BOOTSTRAP INTERVALS (SEC. III-D). BOLD: CONFIRMED PRE-REGISTERED HYPOTHESES. TIME: RELATIVE TO B. †TRAINED ON 5,598 FILES (SEC. III-A), VS. B ON THE SAME SPLIT (TEST $\mathrm{rel}_{\mathrm{all}}$ 2.90%); ‡VALIDATION ONLY.

	Block (position)	Δparams	Time	Seeds	Val. $\Delta\mathrm{rel}_{v_y}$	Test $\Delta\mathrm{rel}_{\mathrm{all}}$	Test $\Delta\mathrm{rel}_{v_y}$
$\mathrm{M}_8^{\mathrm{sh}}$	shared-weight SS2D (enc. 8×8)	+116,736	1.2×	8	-0.24 $[-0.46, -0.04]$	$+0.16$ $[-0.10, +0.53]$	-0.04 $[-0.59, +0.49]$
M_8	SS2D (enc. 8×8)	+169,216	1.2×	8	-0.09 $[-0.35, +0.16]$	-0.01 $[-0.34, +0.26]$	$+0.00$ $[-0.52, +0.50]$
B-large	none, 33 base channels	+147,720	1.3×	8	-0.00 $[-0.20, +0.17]$	–	–
DR-M†	deep path → 11 SS2D at 8×8	+22,656	2.9×	5	-0.37 $[-0.86, +0.14]$	-0.12 $[-0.51, +0.29]$	-0.69 $[-2.12, +0.24]$
TR†	deep path removed	−1,838,720	0.8×	5	–	$+0.57$ $[-0.01, +1.03]$	–
M_{64}	SS2D (enc. 64×64)	+20,800	3.7×	8	-0.75 $[-1.03, -0.50]$	-0.17 $[-0.46, +0.07]$	$\mathbf{-1.17}$ $[-2.47, -0.43]$
$\mathrm{M}_{64}^{\mathrm{loc}}$	SS2D without scan (enc. 64×64)	+7,232	1.6×	8	-0.67 $[-0.98, -0.37]$	-0.12 $[-0.35, +0.15]$	$\mathbf{-0.87}$ $[-1.60, -0.35]$
SA_{64}‡	self-attention (enc. 64×64)	+4,288	17×	5	-0.61 $[-0.86, -0.39]$	–	–
Conv_{64}‡	two 3×3 convs (enc. 64×64)	+6,956	1.2×	5	-0.12 $[-0.32, +0.07]$	–	–
$\mathrm{M}_{64}^{\mathrm{dec}}$‡	SS2D (dec. 64×64)	+20,800	3.7×	5	-0.30 $[-0.53, -0.10]$	–	–
M_{16}‡	SS2D (enc. 16×16)	+55,936	1.3×	5	-0.18 $[-0.42, +0.05]$	–	–
5 variants‡	$\mathrm{M}_{64}^{\mathrm{loc}}$ minus one ingredient	3.6k–43.5k	1.3–1.6×	5	-0.59 to -0.75, all < 0	–	–

as the 8×8 blocks without a new block. Additional variants (32×32 insertion, multi-block placement) yield consistent findings.

D. Protocol and statistics

All models follow the repository's training protocol for a fair comparison: Adam optimizer ($\mathrm{lr} = 6 \times 10^{-4}$, $\beta = (0.5, 0.999)$), L1 loss, batch size 10, evaluation of the final checkpoint. The learning rate is constant for the first half of training, then decays towards $0.1\,\mathrm{lr}$; as this schedule is defined in epochs, we fix 133 epochs (79,667 iterations), the longest of three budgets (17, 67, 133) whose validation gain over the previous one exceeded the seed-to-seed noise, a rule set before the runs. To account for non-deterministic CUDA kernels, we used 8 seeds for baseline B and primary models, 5 for ablations and DR-M/TR, and 3 for the full dataset.

Performance comparisons use the sample-wise mean of seed-averaged differences relative to B, with 95% confidence intervals (CI) from a two-level bootstrap [14]: each of 10,000 repetitions resamples the seeds of each configuration (training variability) and the airfoils with all their samples (flow conditions of one airfoil are correlated). A CI containing zero indicates "no demonstrated difference" rather than equivalence.

Following exploratory validation, test evaluations were pre-registered (specifying hypothesis, metric and the criterion "confirmed if the upper bound of the 95% CI of the difference to B is below zero") and executed exactly once across five passes: (1) $8 \times 8/32 \times 32$ blocks and B-large on $\mathrm{rel}_{\mathrm{all}}$; (2) M_8; (3) DR-M with TR as control; (4) M_{64}; and (5) $\mathrm{M}_{64}^{\mathrm{loc}}$ on rel_{v_y} (hypotheses 4–5 derived from validation, with 5 dependent on 4). Most experiments ran locally on an RTX 4060 Laptop GPU (8 GB), while the full dataset (≈20 GiB) used a cloud VM with an NVIDIA L4 GPU.

IV. RESULTS AND DISCUSSION

A. Mamba is redundant, yet replaces the deep path

On the test set, inserting a Mamba block at 8×8 yields no performance improvement (Table I): $\mathrm{M}_8^{\mathrm{sh}}$ shows a $\mathrm{rel}_{\mathrm{all}}$ difference to B of $+0.16$ $[-0.10, +0.53]$, whose lower bound rules out any reduction larger than 0.10 points (3.6% of B's error), while M_8 yields -0.01 $[-0.34, +0.26]$. These blocks add capacity rather than useful representations: on the training set, M_8 achieves a 16% lower L1 loss than B without improving validation L1, while B-large fails to improve training fit.

As hypothesized (Sec. II-B), the deep path provides sufficient global receptive field. On the filtered split (Sec. III-A), truncating it (TR) is clearly worse on validation ($\mathrm{rel}_{\mathrm{all}}$ $+0.73$ $[+0.58, +0.89]$), while on the test set the interval touches zero ($+0.57$ $[-0.01, +1.03]$). Replacing the pruned layers with 11 SS2D blocks (DR-M) recovers the loss (DR-M−TR -0.69 $[-1.10, -0.26]$ on test) and is not distinguishable from B (-0.12 $[-0.51, +0.29]$). Hence, while global context is necessary and SSM blocks can provide it, the U-Net already captures it (though TR also removes 79% of the parameters).

B. The gain is at the highest resolution

Moving Mamba blocks to higher encoder resolutions progressively improves validation rel_{v_y} (Fig. 2a), from 5.98% at 8×8 to 5.32% at 64×64 (vs. baseline 6.07%); in the decoder the block helps less. On the test set, M_{64} confirms the pre-registered hypothesis, reducing rel_{v_y} by -1.17 $[-2.47, -0.43]$ points (18% relative, with 87% of the airfoils improving) and lowering the aggregate ratio (-0.55 $[-1.19, -0.01]$) and, almost demonstrably, the median sample (-0.50 $[-1.25, +0.03]$), while $\mathrm{rel}_{\mathrm{all}}$, rel_{v_x} and rel_p show no demonstrated difference. This comes at 3.7× the training time.

C. Ablation: the gain does not need the SSM

Fig. 2b shows that removing the spatial scan ($\mathrm{M}_{64}^{\mathrm{loc}}$) retains most validation gains (-0.67 $[-0.98, -0.37]$ vs. B; $+0.08$ $[-0.07, +0.25]$ vs. M_{64}), indicating long-range mixing is not strictly required. The gain is not specific to Mamba: self-attention (SA_{64}) is not distinguishable from M_{64} ($+0.13$ $[-0.07, +0.35]$), but requires 17× the training time. Not every block helps, though: Conv_{64} shows no demonstrated gain over B (-0.12 $[-0.32, +0.07]$) and is worse than $\mathrm{M}_{64}^{\mathrm{loc}}$ ($+0.54$

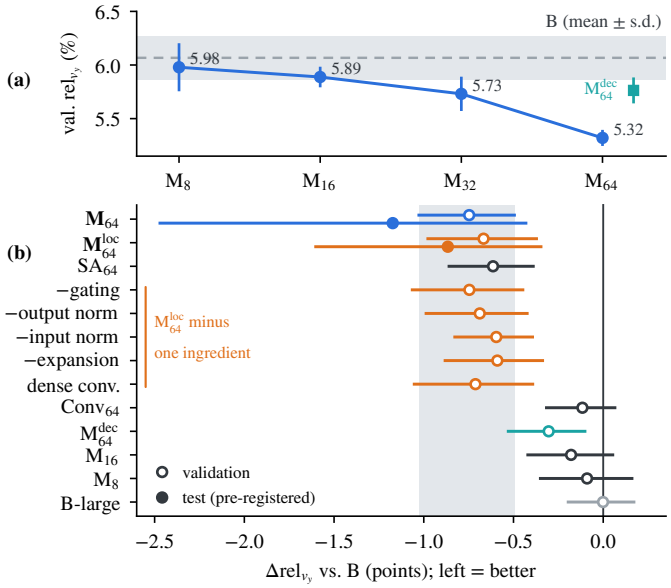


Fig. 2. (a) Validation rel_{v_y} (mean ± s.d. over 8, 5, 3, 8 seeds) vs. resolution of the encoder block; square: same block in the decoder; band: B. (b) $\Delta\mathrm{rel}_{v_y}$ vs. B with 95% CIs. Open: validation (8 seeds for M_{64}, $\mathrm{M}^{\mathrm{loc}}_{64}$, M_8, B-large; 5 for the others); filled: pre-registered test. Band: validation CI of M_{64}.

$[+0.24, +0.89]$). Removing one component of $\mathrm{M}^{\mathrm{loc}}_{64}$ at a time never loses the gain (all variants below B, none distinguishable from $\mathrm{M}^{\mathrm{loc}}_{64}$): the components may be partly redundant, or the gain may come from what they share and Conv_{64} lacks (pointwise projections, SiLU, per-channel scale); our data do not separate the two.

Our final architecture combines B with $\mathrm{M}^{\mathrm{loc}}_{64}$ (+7,232 parameters, +7% FLOPs, 1.6× training time). On the test set, its pre-registered hypothesis is confirmed: rel_{v_y} drops by -0.87 points $[-1.60, -0.35]$ (13% relative reduction), with 90% of airfoils improving. This is about three quarters of the gain of M_{64} (-0.87 vs. -1.17); the direct comparison $\mathrm{M}^{\mathrm{loc}}_{64}-\mathrm{M}_{64}$ ($+0.31$ $[-0.15, +1.11]$) shows no demonstrated difference in either direction, so the contribution of the SSM remains undetermined. Unlike MambaOut's dense tasks [2], spatial recurrence is not shown to be required here.

D. Expected on pressure, found on v_y

We expected gains on p (Eq. (1)) and found them on v_y. Our untested reading is that the smooth, long-range part of p is already captured by the U-Net, which works like a multigrid V-cycle [15], while residual p errors concentrate in severe separation tails missed by all models (2.5% of validation samples account for 15% of total rel_p). On the test set the mean rel_p decreases for both blocks (M_{64} -1.79 $[-5.40, +0.09]$, $\mathrm{M}^{\mathrm{loc}}_{64}$ -1.37 $[-3.38, -0.12]$), but for M_{64} the median sample does not move (-0.01 $[-1.43, +0.92]$): the change sits in a few samples, and p was not a pre-registered hypothesis.

The gains are instead local: within a 4-pixel wall band, the v_y error drops by 11% (M_{64}) and 10% ($\mathrm{M}^{\mathrm{loc}}_{64}$). Since v_y is mostly induced by the airfoil geometry, this improvement shows in rel_{v_y}, while the freestream dilutes it in v_x and $\mathrm{rel}_{\mathrm{all}}$.

E. More data and limitations

On the full dataset (53,830 samples, including sheared airfoils; validation only, 3 seeds, after the test passes) all models improve markedly and the gain halves: $\mathrm{M}^{\mathrm{loc}}_{64}$ achieves a difference to B of -0.22 $[-0.41, -0.04]$ on regular validation files (-4.7% vs. -8.2% on the reduced set, same seeds), concentrated near the wall (wall-band v_y MAE -6%), with no demonstrated change in the aggregate ratio.

Finally, key limitations include a test suite of 30 airfoils, five test passes, a single optimizer for all architectures, and separated flows at high angles, which every model misses and whose steady RANS targets may not be fully converged. Addressing these limitations is left for future work.

V. Conclusion

Mamba blocks bring no benefit at deep U-Net levels, where global context is already established. Instead, at high resolution they reduce the v_y error by 18% on the test set, a gain concentrated near the airfoil and mostly due to the local block design: without the SSM ($\mathrm{M}^{\mathrm{loc}}_{64}$) it becomes 13% at 1.6× the training time, consistent with MambaOut; eight times more data halves it.

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